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**1** The temperature at 3 am is −9 °C.
The temperature at 2 pm is 17 °C.

Find the increase in temperature from 3 am to 2 pm.

..... °C [1]

**2** Find **all** the common factors of 18 and 30.

..... [1]

**3** A cuboid has length 8 cm, width 4 cm and height 3.5 cm.

Work out the volume of the cuboid.

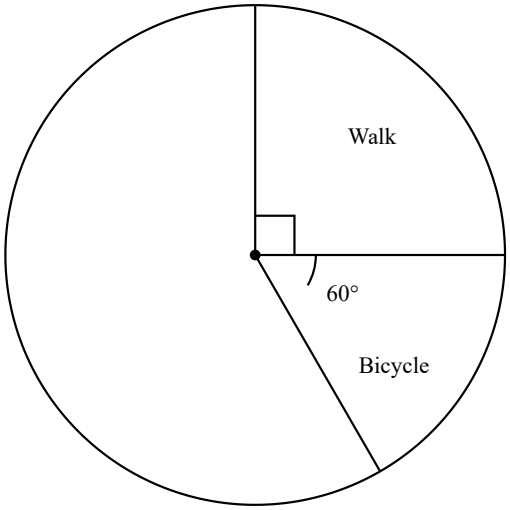
..... $cm^3$ [2]

**4** A tablet costs $480.
In a sale, this cost is reduced by 15%.

Find the cost of the tablet in the sale.

$ ..... [2]

**5** The pie chart shows some information about the way 720 students travel to school.




**(a)** Work out the number of students that walk to school.

..... [2]

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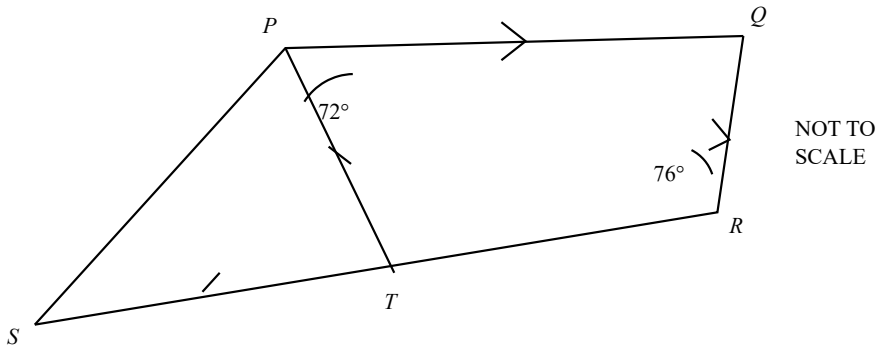
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**(b)** 144 of the students travel to school by car.
The remaining students travel by bus.

Complete the pie chart.

[3]

**6**




*PQRS* is a quadrilateral.
*T* lies on *RS* and *PT* is parallel to *QR*.
*PT* = *TS*.

Find

**(a)** angle *PQR*

Angle *PQR* = ..... [1]

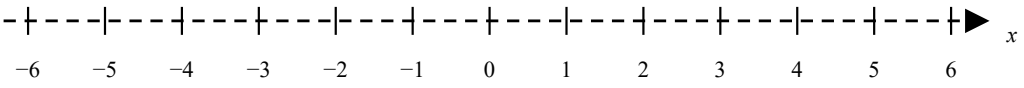
**(b)** angle *PTS*

Angle *PTS* = ..... [1]

**(c)** angle *SPQ*.

Angle *SPQ* = ..... [2]

**7** Represent the inequality $-5 < x \leq 2$ on the number line.




[2]

**8** Tomas buys *n* notebooks that each cost 65 cents.
He pays with $10.

Write an expression, in terms of *n*, for the change, in **cents**, Tomas receives from the $10.

..... cents [2]

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**9** Priya has a full bottle of juice.

She drinks $\frac{2}{5}$ of the full bottle on Monday.

She drinks $\frac{1}{4}$ of the full bottle on Tuesday.

Find the fraction of the bottle of juice remaining.

.....

[3]

**10**

$$T = pq + 3qr$$

**(a)** $p = 2.97$, $q = 8.13$ and $r = 9.84$ .

By rounding each value correct to 1 significant figure, work out an estimate for $T$.

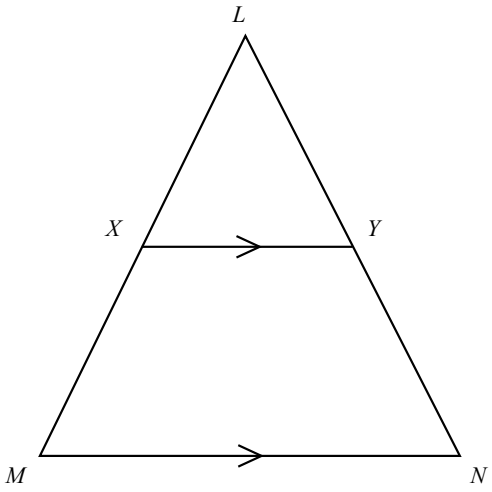
.....

[3]

**(b)** Rearrange the formula to make $q$ the subject.

$q =$ ..... [2]

**11**




In the diagram, $X$ lies on $LM$ and $Y$ lies on $LN$.
$XY$ is parallel to $MN$.

**(a)** Explain why triangle $LMN$ is mathematically similar to triangle $LXY$.
Give a geometrical reason for each statement you make.

.....

.....

.....

.....

[3]

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**(b)** $XY = 4$ cm, $MN = 12$ cm and $LX = 6$ cm.

Work out $LM$.

$LM =$ ..... cm [2]

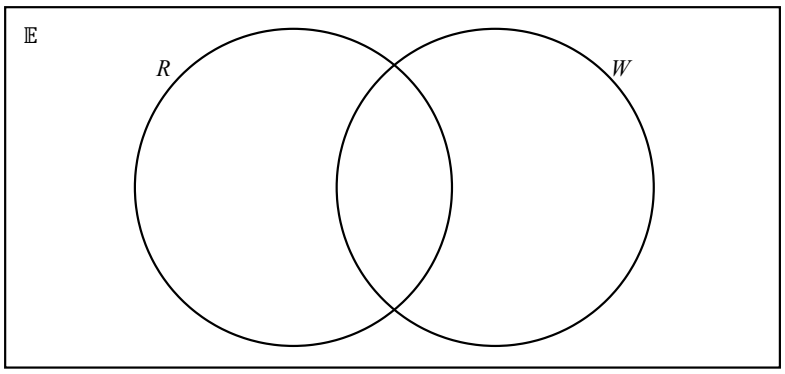
**(c)** The area of triangle $LXY$ is $3k$ cm$^2$.

Find, in terms of $k$, the area of quadrilateral $MNYX$.

..... cm$^2$ [2]

**12** A sports club has 120 members.

75 run ($R$).
80 swim ($W$).
20 do not run or swim.




**(a)** Complete the Venn diagram.

[3]

**(b)** One member of the sports club is chosen at random.

For this member, find

**(i)** P($W$)

.....

[1]

**(ii)** P($R \cap W$)

.....

[1]

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**(iii)** $P(R \cup W')$.

.....  [1]

**13**

$$N = 2^5 \times 3^4 \times 7^2$$

**(a)** Write $12N$ as a product of its prime factors.
Give your answer in index form.

.....  [2]

**(b)** $R$ is an integer.
$\frac{N}{R}$ is a cube number.

Find the smallest possible value of $R$.

$R =$ .....  [2]

**14** Find the value of

**(a)** $4^3 \div 4^{-1}$

.....  [2]

**(b)** $8^{-2/3}$.

.....  [2]

**15** Factorise.

**(a)** $x^2 - 121$

.....  [1]

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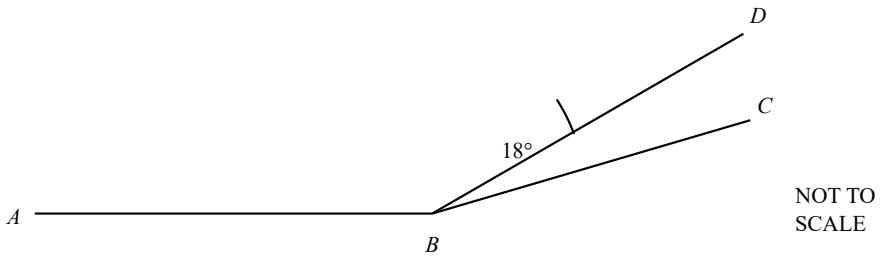
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**(b)** $7a(a - 3b) + 4(a - 3b)^2$

.....

[2]

**16**




*AB* and *BD* are two sides of a regular 20-sided polygon.
*AB* and *BC* are two sides of a regular *n*-sided polygon.
Angle *DBC* = 18°.

Work out the value of *n*.

*n* = .....

[4]

**17** $B$ is the point $(-2, 3)$ and $D$ is the point $(-6, 9)$.
$BD$ is a diagonal of the kite $ABCD$.

**(a)** The ratio of the lengths of the diagonals $BD : AC = 2 : 5$.

Work out the length of $AC$.
Give your answer as a surd in its simplest form.

..... [5]

**(b)** Find the coordinates of the midpoint of $BD$.

( ....., ..... ) [2]

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**(c)** The diagonal $AC$ of the kite passes through the midpoint of $BD$.

Find an equation of $AC$.
Give your answer in the form $y = mx + c$.

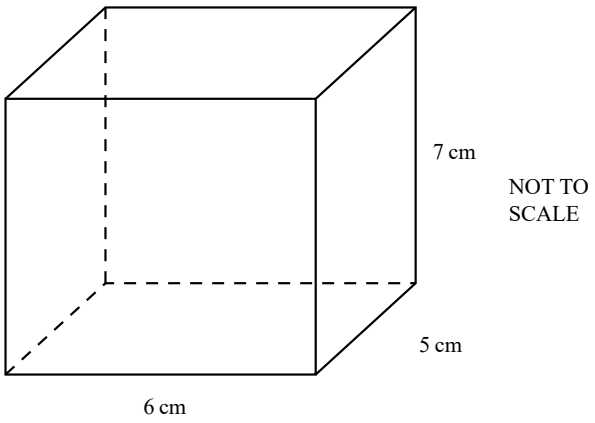
$y =$ ..... [4]

**18** Rationalise the denominator and simplify.

$$\frac{18}{5 + \sqrt{7}}$$

..... [3]

**19**




The diagram shows a box in the shape of a cuboid.
Mala has a straight rod of length 12 cm.

Show that this rod does **not** fit completely inside the box.

[3]

**20**

$$f(x) = \frac{5}{2x-1}, x \neq \frac{1}{2} \qquad g(x) = x + 4$$

**(a)** Find

**(i)** g(2)

..... [1]

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**(ii)** gf(3)

..... [2]

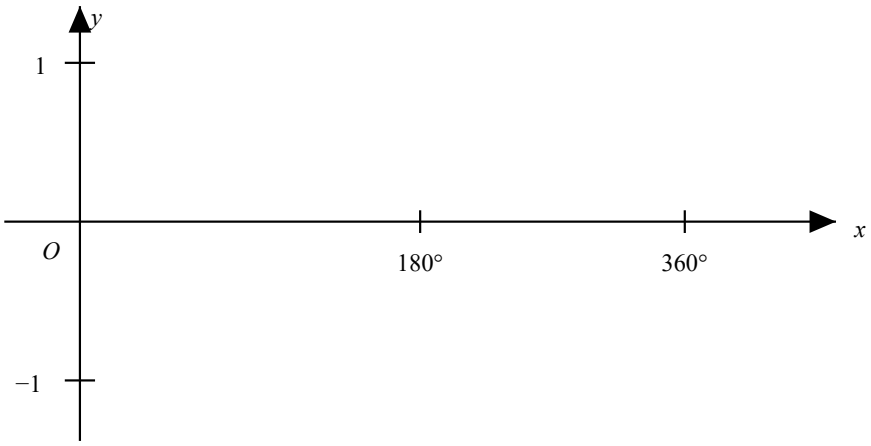
**(iii)** $f^{-1}(x)$.

$f^{-1}(x) =$ ..... [3]

**(b)** Solve $f(x) = g(x)$.

$x =$ ..... or $x =$ ..... [5]

**21** **(a)**




On the diagram, sketch the graph of $y = \sin x$ for $0° \leq x \leq 360°$. [2]

**(b)** Solve the equation $2 \sin x + 1 = 0$ for $0° \leq x \leq 360°$.

$x =$ ..... or $x =$ ..... [3]

**22** A graph with equation $y = x^2 + bx + c$ has a minimum point at $(-4, 7)$.

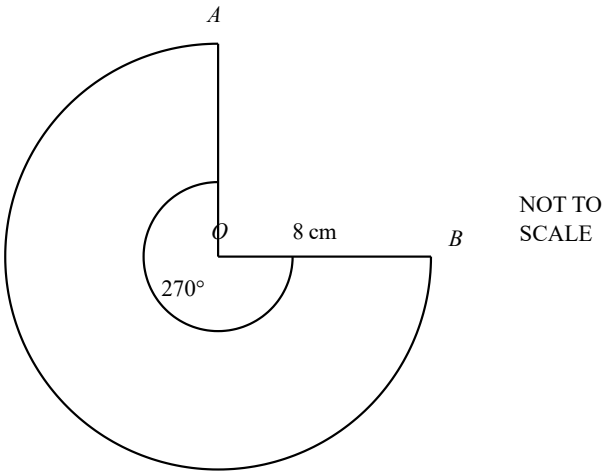
Find the value of $b$ and the value of $c$.

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$b =$ .....

$c =$ ..... [3]

**23**




The diagram shows a major sector, $AOB$, of a circle.
The sector angle is 270° and the radius is 8 cm.

**(a)** Show that the length of the major arc $AB$ is $12\pi$ cm.

[1]

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**(b)** *OA* is joined to *OB* to form a cone.

Work out the volume of the cone.
Give your answer in the form $\frac{(a\sqrt{b})\pi}{3}$ where $a$ is an integer and $b$ is a prime number.

..... cm$^3$ [6]

**24**

$$8^{nx} = (4^x)^3$$

Find the value of $n$.

$n =$ ..... [2]

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# Answer key

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| | |
|---|---|
| **1** | 26 °C |
| **2** | 1, 2, 3, 6 |
| **3** | $8 \times 4 \times 3.5 = 112$ cm$^3$ |
| **4** | $480 \times 0.85 = \$408$ |
| **5a** | $90/360 \times 720 = 180$ students |
| **5b** | car $144/720 \times 360 = 72°$; bus $= 720 - 180 - 120 - 144 = 276 \rightarrow 138°$ |
| **6a** | $180 - 72 = 108°$ (co-interior angles, $PT \parallel QR$) |
| **6b** | 76° (corresponding angles) |
| **6c** | triangle $PTS$ isosceles → angle $TPS = (180 - 76)/2 = 52°$; angle $SPQ = 52 + 72 = 124°$ |
| **7** | open circle at −5, closed circle at 2, joined |
| **8** | $1000 - 65n$ cents |
| **9** | $1 - 2/5 - 1/4 = 1 - 8/20 - 5/20 = 7/20$ |
| **10a** | $3, 8, 10 \rightarrow 3(8) + 3(8)(10) = 24 + 240 = 264$ |
| **10b** | $T = q(p + 3r) \rightarrow q = T / (p + 3r)$ |
| **11a** | angle $L$ common; angle $LXY$ = angle $LMN$ (corresponding angles, $XY \parallel MN$); third angles equal → similar (AAA) |
| **11b** | scale factor $12/4 = 3$, so $LM = 3 \times 6 = 18$ cm |
| **11c** | area $LMN = 9 \times 3k = 27k$; $MNYX = 27k - 3k = 24k$ cm$^2$ |
| **12a** | both $= 75 + 80 - 100 = 55$; only $R = 20$, only $W = 25$, outside $= 20$ |
| **12b i** | $80/120 = 2/3$ |
| **12b ii** | $55/120 = 11/24$ |
| **12b iii** | $75 + 40 - 20 = 95 \rightarrow 95/120 = 19/24$ |
| **13a** | $12N = 2^2 \times 3 \times 2^5 \times 3^4 \times 7^2 = 2^7 \times 3^5 \times 7^2$ |
| **13b** | need exponents ÷ 3: $R = 2^2 \times 3 \times 7^2 = 588$ ($N/R = 2^3 3^3 = 216 = 6^3$) |
| **14a** | $4^4 = 256$ |
| **14b** | $1 / (\sqrt[3]{8})^2 = 1/4$ |
| **15a** | $(x - 11)(x + 11)$ |
| **15b** | $(a - 3b)(11a - 12b)$ |
| **16** | interior angle of 20-gon = 162°; angle $ABC = 162 - 18 = 144°$; $360/n = 36 \rightarrow n = 10$ |
| **17a** | $BD = \sqrt{(16 + 36)} = 2\sqrt{13}$; $AC = 5/2 \times 2\sqrt{13} = 5\sqrt{13}$ |
| **17b** | (−4, 6) |
| **17c** | gradient $BD = -3/2$, so gradient $AC = 2/3$; $y = (2/3)x + 26/3$ |

| | |
|---|---|
| **18** | $18(5 - \sqrt{7}) / (25 - 7) = 18(5 - \sqrt{7})/18 = 5 - \sqrt{7}$ |
| **19** | diagonal$^2$ = 36 + 36 + 64 = 136; $12^2 = 144 > 136$, so the diagonal is less than 12 cm → the rod does not fit |
| **20a i** | 2 + 4 = 6 |
| **20a ii** | f(3) = 5/5 = 1; g(1) = 5 |
| **20a iii** | $(x + 5) / (2x)$ |
| **20b** | $5 = (x+4)(2x-1) \rightarrow 2x^2 + 7x - 9 = 0 \rightarrow (2x + 9)(x - 1) = 0 \rightarrow x = 1$ or $x = -4.5$ |
| **21a** | sine curve: 0 at 0°, max 1 at 90°, 0 at 180°, min −1 at 270°, 0 at 360° |
| **21b** | $\sin x = -0.5 \rightarrow x = 210°$ or $x = 330°$ |
| **22** | $y = (x + 4)^2 + 7 = x^2 + 8x + 23 \rightarrow b = 8, c = 23$ |
| **23a** | $(270/360) \times 2\pi \times 8 = (3/4)(16\pi) = 12\pi$ cm |
| **23b** | $2\pi r = 12\pi \rightarrow r = 6$; $l = 8$; $h = \sqrt{(64 - 36)} = 2\sqrt{7}$; $V = \frac{1}{3}\pi(36)(2\sqrt{7}) = 24\sqrt{7}\pi = (72\sqrt{7})\pi/3$ |
| **24** | $2^{3nx} = 2^{6x} \rightarrow 3n = 6 \rightarrow n = 2$ |

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